NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

SC4001 Neural Networks and Deep Learning

Deep Learning for ECG Heartbeat Classification

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1. Introduction

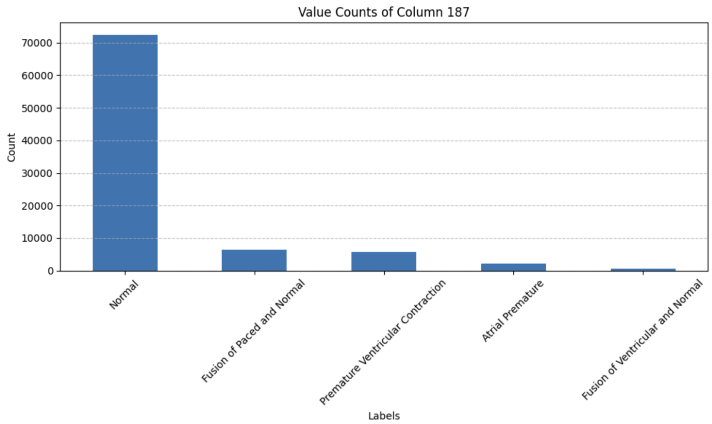
Accurate electrocardiogram (ECG) classification is key to diagnosing cardiovascular diseases and guiding timely medical interventions. This report presents our findings after exploring various neural network architectures to categorise ECG heartbeat signals. We discuss the strengths and weaknesses of each model architecture, as well as analyse errors and potential causes. We aim to explore and compare a range of neural network models, including both base architectures (such as GRU, LSTM, and Attention models) and hybrid architectures that combine Convolutional Neural Networks (CNNs) with these base models.

Our project ultimately seeks to address the following questions in the context of ECG classification and analysis:

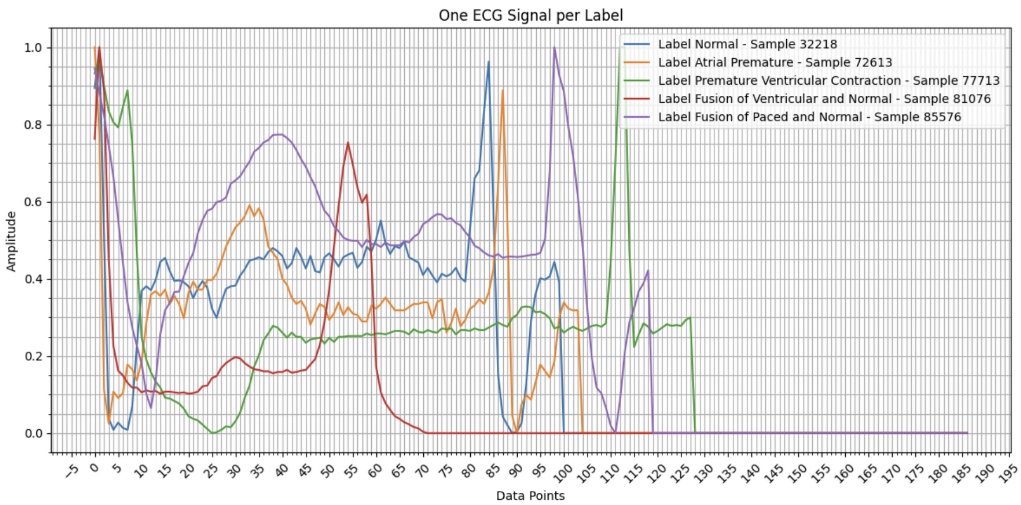
- Which neural network architecture achieves the highest classification performance?
- What are the differences in prediction errors between base models and hybrid models?
- What effect does data augmentation have on model performance?

2. Exploratory Data Analysis

This dataset is composed of two collections of heartbeat signals derived from two famous datasets in heartbeat classification. We decided to work on the MIT-BIH Arrhythmia Dataset which consist of 5 classes: ['N': 0, 'S': 1, 'V': 2, 'F': 3, 'Q': 4]



We chose to sample one ECG signal per label, ensuring each is standardised to a fixed length of 187 data points, either by cropping or padding with zeros. This preprocessing step guarantees consistent dimensions across all samples, which is essential for machine learning models that require uniform input sizes. Since the samples are already of fixed length and scale, no further preprocessing for length adjustment is needed.



3. Model Selection

The accuracy score and F1-score were used to evaluate the performance of each model. In multi-class classification, the F1-score is a suitable metric as it is often interpreted as the harmonic mean between precision and recall. It is computed by the formula:

$$F1 = \frac{2*TP}{2*TP + FP + FN}$$

We will evaluate model performance based on the **weighted-averaged F1-score**, which calculates the F1-score for each class and finds the average weighted by the number of true instances in each class. This accounts for the class imbalance found in our dataset, giving more credit to classes which have higher support.

3.1. Base Models

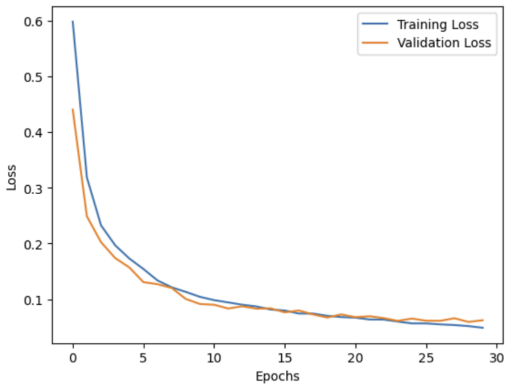
3.1.1. Gated Recurrent Units (GRU)

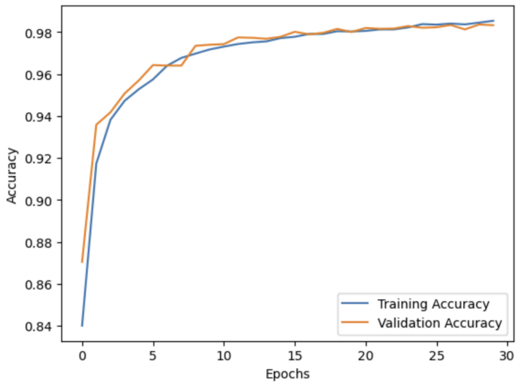
GRUs are a type of Recurrent Neural Network (RNN) which is a class of models that are well-suited in classifying time-series data (Singh et al., 2018). They consist of two gates which are used to control how much of previous inputs should be forgotten or passed into the future. This allows GRUs to model temporal dependencies, as they take both new and past data into consideration.

We built our model by wrapping GRU, Dropout and Dense layers in a Sequential model from the Keras library. Training was conducted in batches of size 64 over 30 epochs, implementing early stopping with a patience of 5 and the Adam optimizer. Categorical cross-entropy was used as the loss function. The architecture selected for model training is shown below.

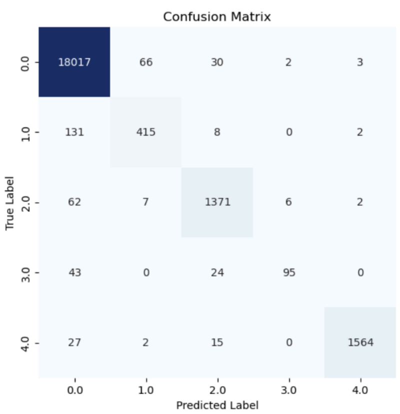
<u>Architecture Description</u>

- Input layer of 187 neurons.
- Two hidden GRU layers of 64 neurons each, followed by a dropout rate of 0.2.
- Dense layer of 5 neurons with softmax activation to generate the probabilities for each class.





By plotting the curves for both training and validation data, we observed that the model learns quickly, reaching a plateau after about 10 epochs. The accuracy then increases steadily until the end of training without triggering early stopping. This suggests that the model could benefit from training on more epochs, although the marginal improvements in accuracy and loss may be very slight. The final accuracy and F1-score obtained on the test data were 98.04% and 97.97% respectively, showing that the GRU model generalises well to unseen data.



The confusion matrix shows that the vast majority of predictions fall within class 0, likely due to the class imbalance in the dataset. It performs exceptionally well on classes 0, 2 and 4, but predicts a significant number of false negatives for the classes with lower support. This indicates that the model struggles to classify ECG signals belonging to under-represented classes, which is crucial in clinical settings as it is desirable to detect rare arrhythmias. Performance could be further improved through data augmentation or conducting resampling methods on specific classes.

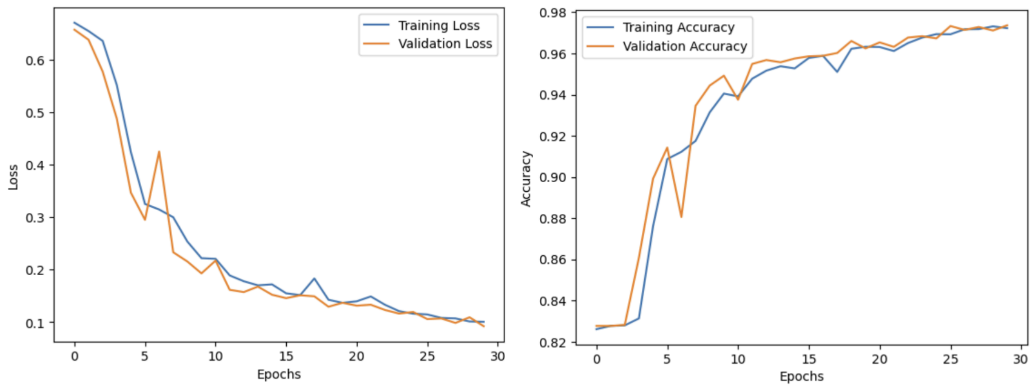
3.1.2. Long Short-Term Memory (LSTM)

LSTMs also belong to the class of RNNs, although they are slightly more complex than GRUs. The main difference is that LSTMs have three gates which control the flow of information, as well as an internal cell state to store past inputs. This means that they are able to learn longer-term dependencies, generally leading to increased accuracy but at the cost of being more computationally expensive due to having more parameters (Hamad, 2023).

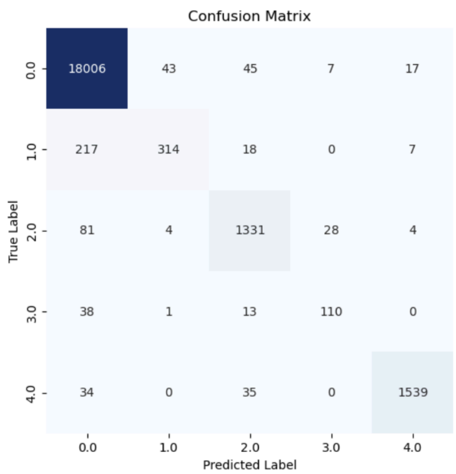
Similarly to Section 3.1.1., we built our LSTM model using Keras. To compare the performance between GRU and LSTM, the number of hidden neurons, dropout rate, batch size and other hyperparameters remained the same. We then chose the following LSTM architecture for training.

<u>Architecture Description</u>

- Input layer of 187 neurons.
- Two hidden LSTM layers of 64 neurons each, followed by a dropout rate of 0.2.
- Dense layer of 5 neurons with softmax activation to generate the probabilities for each class.



The learning curves, especially for validation accuracy and loss, displayed greater fluctuations than that of the GRU model. This may be caused by increased model complexity and hence, more training data may be required. However, the validation curves retain a similar trend to the training curves and appear to stabilise with a greater number of epochs, suggesting that the model could be improved by running more iterations. The LSTM model achieved an accuracy of 97.30% and an F1-score of 97.15% on the test data.

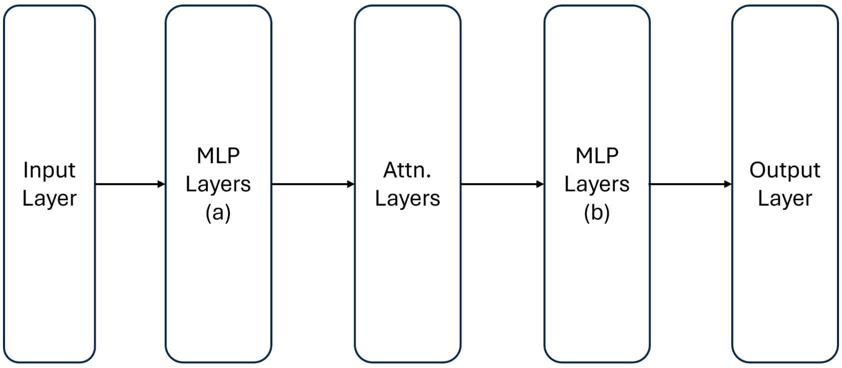


We observed that the LSTM model performed better than the GRU model in predicting true positives in Class 3, indicating a greater ability to learn the patterns of uncommon heart conditions. However, the number of false positives is slightly higher in every other class. When choosing which model to use in real-world applications, it is important to consider the trade-offs in accurately predicting selected classes.

The runtime of the model training for LSTM was comparable to the GRU model despite having more parameters to learn. Therefore, in situations where timely diagnosis is required, the GRU model might be preferred given its stability and simplicity in architecture.

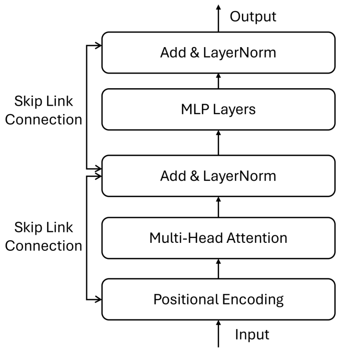
3.1.3. Attention

Given the sequential nature of ECG signal data, attention modelling is expected to be well-suited for the task of heartbeat classification (Vaswani et al., 2017). We constructed models consisting solely of attention layers and multi-layer perceptrons. Training was conducted using early stopping with a patience value of 5, the Adam optimizer, and cross-entropy as the loss function. The following illustrates our high-level network architecture, composed of only Multi-Layer Perceptron and Attention layers.



Architecture Description

- The input layer consists of 187 neurons to accommodate the input sequence length of the ECG dataset.
- The MLP Layers (a) consist of multiple layers of MLP each with 64 neurons, ReLU activation function and a dropout rate of 0.1.
- The Attn. Layers, shown below, each with 4 heads with their MLP layers consisting of a 256-neuron layer and a 64-neuron layer, both with ReLU activation.



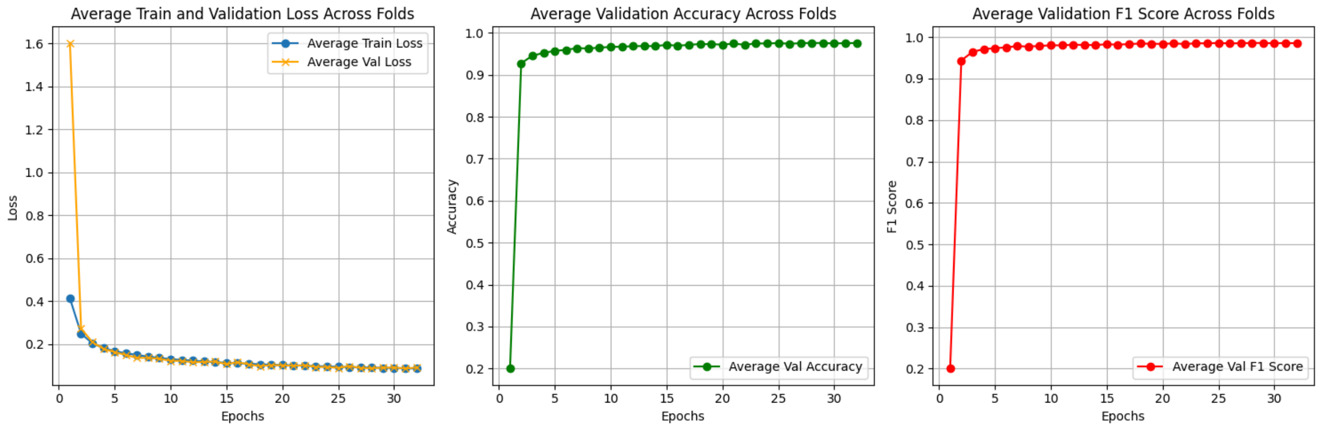
- The MLP Layers (b) first apply mean pooling to achieve a shape of (batch_size, 64), followed by a 64-neuron layer with ReLU activation and a dropout rate of 0.1.
- The output layer consists of 5 neurons, corresponding to the classification classes.

The following table presents slight variations of the proposed architecture and their corresponding F1 scores on the test set.

Variations in model architecture	F1 Score
MLP (a): 2 Layers with 64 neurons each, Attn. Layer: 1 Layer	0.9733
MLP (a): 1 Layer with 64 neurons, Attn. Layer: 1 Layer	0.9756
MLP (a): 4 Layers with 64 neurons each, Attn. Layer: 1 Layer	0.9737
MLP (a): 1 Layer with 64 neurons, Attn. Layer: 2 Layer	0.9738
MLP (a): 1 Layer with 64 neurons, Attn. Layer: 4 Layer	**0.9782**
MLP (a): 1 Layer with 64 neurons, Attn. Layer: 4 Layer, all activation functions: GeLU	0.9730

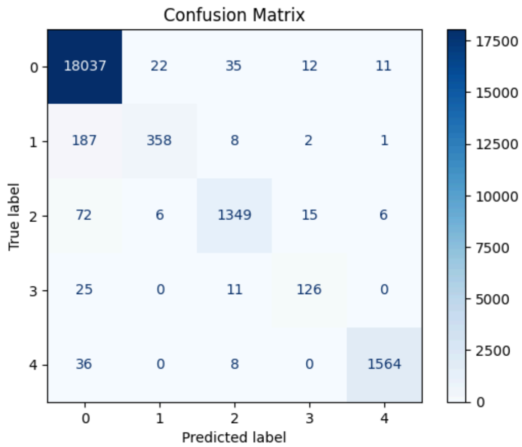
* *Bolded values indicate the best architecture among the experiments.*

Our empirical results indicate no clear trend in performance improvement from increasing the number of MLP layers before the attention block; a single MLP layer in MLP (a) achieved the best performance. However, increasing the number of attention blocks noticeably enhanced model performance. Our best model configuration includes a single MLP layer in MLP (a) and four attention blocks in the attention layers.



** Performance curve reflects the best architecture among the experiments*

We implemented cross-fold validation during training, and the results shown above reflect the average performance of the training and validation sets. We observed a narrow gap between the training loss curve and the validation loss curve, which is ideal and suggests that the model is a good fit. Additionally, the rapid decline in training loss indicates that the model quickly learns the patterns in the ECG signals. This, combined with the narrow gap between training and validation loss, suggests that the model reaches an optimal solution quickly. Overall, this supports the idea that using attention layers in the model architecture is a suitable choice for this task.



Upon reviewing the confusion matrix, we observe that the model's predictions are relatively accurate, with 457 misclassifications out of a total sample size of 21,819, resulting in an approximate classification error of 2.1%. Despite the strong F1-score of 0.9782, the confusion matrix shows that most false negatives occur in Class 0, the majority class. In medical settings, this could be suboptimal, as false negatives may carry a higher risk than false positives (Rowin et. al., 2012).

3.2. Hybrid Models With Convolutional Neural Networks (CNN)

After exploring the base models, we decided to integrate each of them with CNN. CNNs are powerful deep learning tools as each layer in the architecture applies filters to the data to extract a wide range of features. Although traditionally used in image recognition tasks, they have also been used in ECG classification to achieve high overall performance (Xu and Liu, 2020). Therefore, we attempted to combine them with our base models to assess whether prediction performance could be further optimised.

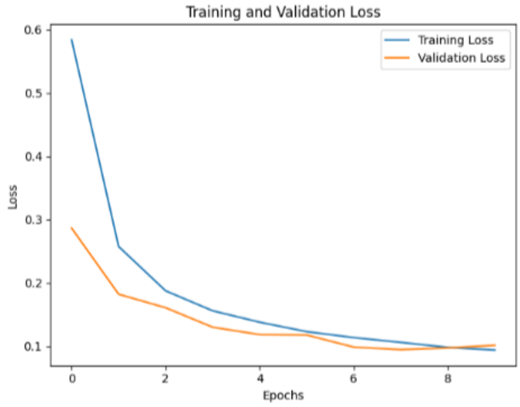
3.2.1. CNN-GRU Model

CNN-GRU model leverages both spatial and temporal features, making it well-suited for ECG classification tasks where both waveform patterns and sequential dependencies are important for accurate diagnosis.

<u>Architecture Description</u>

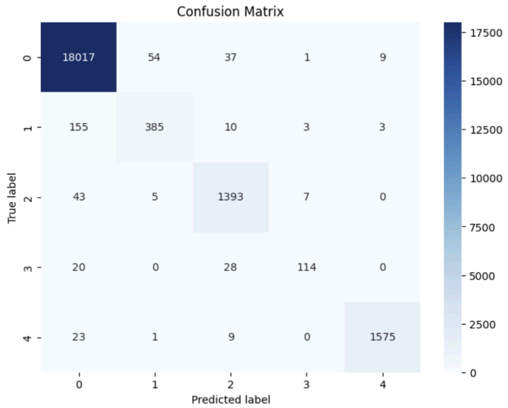
- Input Layer: 187 neurons
- A Conv1D layer: 128 filters is followed by batch normalisation and dropout for regularisation.
- Two GRU layers, where the first GRU returns the entire sequence, and the second GRU outputs a reduced 64-dimensional representation.
- Dense Layers: 1 dense layer with 128 units is followed by dropout, then a final dense layer with softmax activation for classification into 5 classes.

Before implementing early stopping, the CNN-GRU model achieved an accuracy of 97.11%, while these metrics indicate good performance, further improvements were desired to enhance the model's generalisation capability.



While the training loss continues to decrease, the validation loss appears to plateau towards the end, indicating that further training does not improve the model's generalisation performance. This trend suggests that early stopping could be beneficial in preventing overfitting by halting training once the validation loss stops improving.

After implementing early stopping, the model's performance improved significantly. With an overall accuracy of 98.14% on the test set, the model demonstrates effective classification capability for all five ECG labels. The confusion matrix further highlights the model's robustness, showing minimal misclassifications. Notably, the majority class (label 0) has a recall of 99.44%, while other classes achieve competitive recall rates, such as 97.47% for label 4.



Based on the confusion matrix, while Class 0 (Normal) still has the highest number of true positives (18,017), other classes, such as Class 2 and Class 4, also show high accuracy, with minimal misclassifications across neighbouring classes. This suggests that the model is capable of distinguishing between the various ECG classes, which is essential for detecting rare arrhythmias and accurately classifying diverse heartbeat patterns.

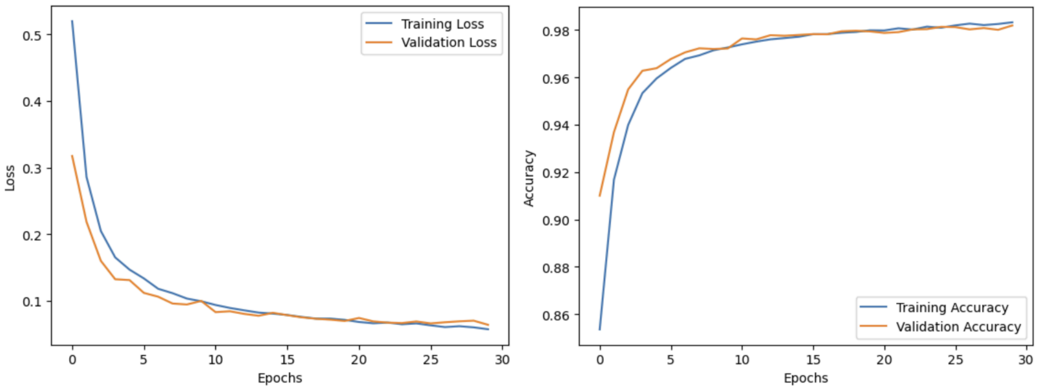
The overall high accuracy of the model with a high weighted average F1-score of 98.07%, reflecting the model's strong general performance. These improvements suggest that early stopping effectively prevented overfitting, allowing the model to generalise better to unseen data.

3.2.2. CNN-LSTM Model

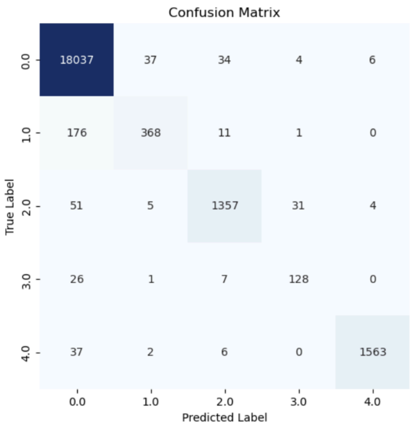
Our second hybrid model combines the architectures of both CNN and LSTM, attempting to leverage the ability of LSTMs to capture long-term temporal dependencies.

<u>Architecture Description</u>

- Input layer of 187 neurons.
- Two 1-dimensional convolutional layers of 64 and 128 neurons respectively, with a ReLU activation function and a kernel size of 3. Each layer is followed by a MaxPooling1D layer with a pool size of 2 and a dropout rate of 0.2.
- One LSTM layer of 64 neurons, followed by a dropout rate of 0.2.
- Dense layer of 5 neurons with softmax activation to generate the probabilities for each class.



The learning curves indicate that the model demonstrates good performance as both training and validation accuracy converge toward similar levels. Furthermore, the validation curve is smooth and shows minimal signs of diverging from the training curve. Overall, we can infer that the model is stable and generalises well, achieving an accuracy and F1-score of 97.99% and 97.92% respectively on the test data, showing improvements from the base LSTM model.

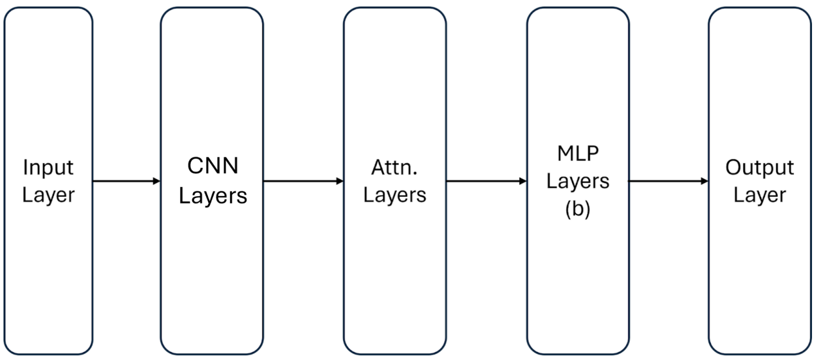


By comparing the confusion matrix with that of the base model, we observed that the number of true positives increased across all classes with a corresponding decrease in false negatives. This supports our hypothesis that the combination of convolutional and recurrent layers enhances the model's ability to identify key features in the ECG signals. As a result, fewer misclassifications were made which leads to improved diagnostic reliability.

We also observed that the training time for the hybrid model was considerably shorter compared to the base model. This combination of higher accuracy, shorter training time and overall robustness suggests that the hybrid model is well-suited for real-time clinical use.

3.2.3. CNN-Attention Model

We also experimented with incorporating CNN into the attention-based model. The selected architecture follows the structure proposed in Section 3.1.3, with the only change being the replacement of MLP Layers (a) with CNN Layers.



Architecture Description

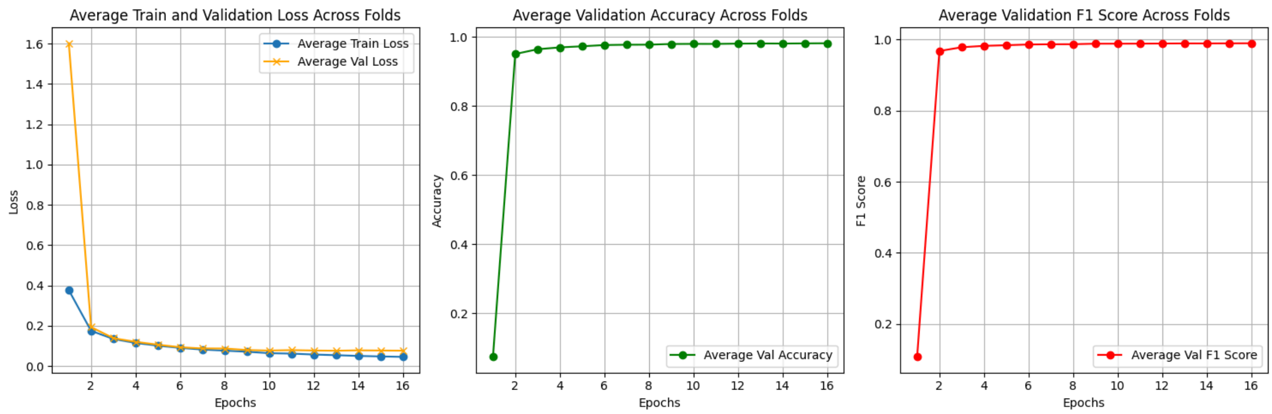
- The input layer consists of 187 neurons to accommodate the input sequence length of the ECG dataset.
- The CNN layers consist of multiple 1D convolutional layers, each with a kernel size of 3, a channel size of 64, and padding size of 1. Each convolutional layer includes a skip connection before and after the Conv1D layer, followed by max pooling and a dropout rate of 0.1. The output is then flattened and passed into a 64-neuron MLP layer.
- The subsequent layers follow the same structure as the base model proposed in Section 3.1.3.

The following table presents slight variations of the proposed architecture and their corresponding F1 scores on the test set.

Variations in model architecture	F1 Score
CNN Layer: 1 Conv1D layer, Attn. Layer: 1 Layer	0.9785
CNN Layer: 4 Conv1D layers, Attn. Layer: 1 Layer	0.9805
CNN Layer: 4 Conv1D layers, Attn. Layer: 4 Layers	**0.9816**

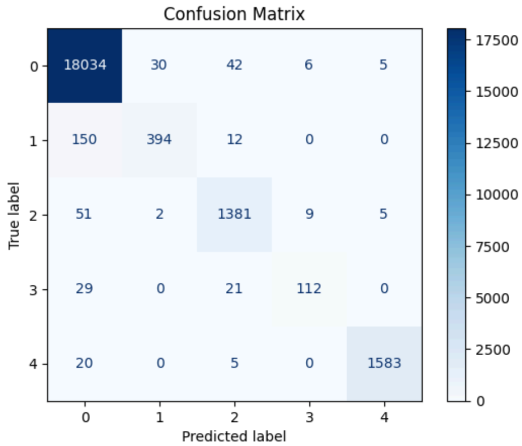
* *Bolded values indicate the best architecture among the experiments.*

As observed from our empirical results, when comparing the base model in Section 3.1.3, which includes 1 MLP layer and 1 attention layer, to a model with 1 Conv1D layer and 1 attention layer, the performance improved from 0.9756 to 0.9785. Adding more CNN layers further improved the model to 0.9805, and incorporating additional attention layers boosted performance to 0.9816.



* *Performance curve reflects the best architecture among the experiments*

The training performance of the hybrid CNN-Attention model exhibits behaviour similar to the base model. An interesting observation is that early stopping is triggered more quickly in the hybrid CNN-Attention model compared to the base model, suggesting more effective learning in the hybrid model. Despite the increase in the number of trainable parameters—from 214K in the base model to 620K in the hybrid model—the training time did not show a significant change for either model, likely due to the faster convergence and the ability to parallelize computations in both models.



Overall, the hybrid model shows a slight improvement over the base model, with the F1-score increasing from 0.9782 to 0.9816 and the classification error reducing to approximately 1.8%, compared to 2.1% in the base model. However, similar behaviour is observed in the hybrid model, where most false negatives occur in Class 0, the majority class.

4. Experimental Analysis

4.1. Evaluating Performance

To evaluate the effectiveness of various deep learning models for ECG heartbeat classification, we compared three base models—GRU, LSTM, and Attention models—with their hybrid counterparts, which incorporate CNN for enhanced feature extraction. We aimed to analyse model performance based on F1-scores, an appropriate metric given the class imbalance in the dataset, as well as the differences in prediction errors and the effect of data augmentation on the overall accuracy and generalisation capability of each model.

Model	**F1-Score**
GRU Base Model	0.9797
LSTM Base Model	0.9715
Attention Base Model	0.9782
CNN-GRU Hybrid Model	0.9814
CNN-LSTM Hybrid Model	0.9792
CNN-Attention Hybrid Model	**0.9816**

Our results indicate that each hybrid model performed marginally better than its respective base model, with CNN layers enhancing feature extraction by capturing localised spatial patterns within the ECG signals. Specifically, the CNN-Attention hybrid model achieved the highest F1-score of 0.9816, surpassing all other models, including the base Attention model, which scored 0.9782. This improvement highlights the advantage of integrating CNN with attention mechanisms, as it combines both spatial and sequential learning capabilities critical for ECG classification.

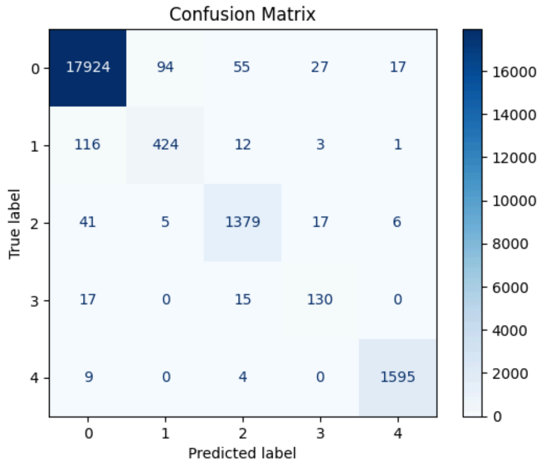
Although the hybrid models performed better, the overall increase in performance was relatively small, and the model still exhibited false negatives, largely due to misrepresentation in the dataset. The class imbalance and under-representation of certain heartbeat types made it challenging for the model to accurately classify these minority classes.

4.2. Analysis of Prediction Errors

We further analysed prediction errors by examining confusion matrices for each model to identify variations in false positives and false negatives, especially in minority classes. The base models, although effective, displayed higher rates of false negatives in these under-represented classes, which could lead to missed detections of certain heart conditions. The hybrid models, particularly those incorporating CNN layers, showed a reduction in false negatives. The CNN-GRU and CNN-Attention models achieved higher recall for these minority classes, indicating that CNN helped capture subtle features crucial for accurate classification of less common heartbeat patterns.

The consistent slight reduction in misclassifications in hybrid models further supports that combining CNN with sequence models improves diagnostic reliability. This could be advantageous particularly in health settings, where minimising false negatives is essential to avoid undiagnosed conditions.

5. Effects of Data Augmentation



Due to the ability for parallel computation, we tested the resampled dataset with the hybrid CNN-Attention model. We upsampled all classes using random sampling with replacement to match the sample size of the majority class (Class 0). However, after upsampling, we observed a slight drop in F1-score to 0.9799, compared to the original hybrid model trained on the unmodified dataset. Interestingly, the confusion matrix reveals that while the number of false negatives for Class 0 has decreased, false positives have increased. Despite the lower F1-score, this model may better suit the dataset's practical use case, particularly in medical settings where false negatives are often more concerning than false positives (Rowin et al., 2012).

Due to the sequential nature of LSTM and GRU models (Hochreiter & Schmidhuber, 1997; Cho et al., 2014) training them on the augmented dataset would be computationally expensive. These models process data step by step, limiting their ability to efficiently parallelize computations. Furthermore, on the original dataset, the hybrid models exhibit similar prediction behaviour to the hybrid CNN-Attention model. Therefore, it is reasonable to assume that their performance on the augmented dataset would be comparable to that of the CNN-Attention model.

6. Conclusion

In this project, we explored various deep learning architectures for ECG heartbeat classification, including base models (GRU, LSTM, Attention) and hybrid models that incorporate CNN layers. The base models performed well on their own, achieving high F1 scores, but the addition of CNN layers in the hybrid models provided a slight improvement in accuracy and generalisation by enhancing feature extraction. The CNN-Attention hybrid model achieved the highest F1 score, demonstrating the advantage of combining spatial and sequential learning for this task.

We also addressed the class imbalance by upsampling minority classes to match the sample size of the majority class. This resulted in a slight decrease in overall F1 score due to an increase in false positives, but it reduced false negatives, particularly for the majority class. This trade-off could be valuable in medical contexts where minimising missed detections (false negatives) is critical.

In conclusion, our study demonstrates the potential of hybrid deep learning architectures, especially the CNN-Attention model, for ECG classification tasks. Additionally, balancing class representation through data augmentation can improve the model's reliability in practical applications. This research highlights the importance of selecting appropriate model architectures and data preprocessing techniques to develop robust models for healthcare applications, where accuracy and error type are both essential factors.

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